

Soumil Malik

+91-9650799486 | soumil4malik@gmail.com | [linkedin/soumilmalik24](https://www.linkedin.com/in/soumilmalik24) | [github/SoumilMalik24](https://github.com/SoumilMalik24)

ABOUT ME

CS undergraduate specializing in Agentic AI and LLMOps — building autonomous multi-agent systems, distributed AI backends, and production RAG pipelines. Proficient across the full GenAI stack: LangGraph, CrewAI, MCP, LangSmith, vector databases, and async task queues. Focused on AI that reasons, acts, and delivers structured outputs beyond simple chat.

EDUCATION

Guru Gobind Singh Indraprastha University <i>B.Tech in Computer Science and Engineering CGPA: 8.39/10.0</i>	Delhi, India <i>Aug. 2022 – May 2026</i>
---	---

EXPERIENCE

AI/ML Engineer Intern <i>RegalRinse</i>	Nov. 2025 – Jan. 2026 <i>Remote</i>
- Engineered a teacher-student distillation pipeline transferring SOTA Qwen-Image-Edit reasoning into a Stable Diffusion 1.5 + LoRA adapter for text-controlled wall restoration and recoloring. - Built a synthetic data factory using a 4-bit quantized teacher model; achieved 10× lower latency and 75% VRAM reduction (16GB → 4GB) enabling free-tier GPU deployment. - Developed a regression-based pricing model estimating house construction costs across varied structural and material configurations.	

PROJECTS

ResearchSynth <i>Live</i> <i>LangGraph, LangSmith, FastAPI, Celery, Redis, Pinecone</i>	Apr. 2026
- Built an autonomous multi-agent research platform using LangGraph with Search, Translation, and Synthesis agents to generate structured markdown research reports with citations from natural language queries. - Engineered an async distributed backend using FastAPI, Celery, Redis, async SQLAlchemy, and PostgreSQL for scalable job execution, persistent history tracking, and real-time research workflow management. - Integrated LangSmith observability for tracing token usage, cost, latency, and agent reasoning; implemented Google OAuth + JWT authentication and cross-lingual research support through a dedicated Translation Agent.	
Smart Finance AI <i>Live</i> <i>FastAPI, LangChain, MCP, OpenAI, React, SSE</i>	Feb. 2026
- Multi-Server MCP platform with isolated microservices for financial math (SIP/EMI/CAGR/Monte Carlo), investment simulation, expense analysis, and inline Base64 chart generation. - FastAPI + LangChain orchestrator with session memory, intelligent tool routing, and real-time SSE token streaming for a ChatGPT-style UX. - Applied production-grade patterns: API key isolation, token trimming, tool caching, structured SSE error events, and multi-session support without a persistent database.	
CodeLens <i>Live</i> <i>CrewAI, LangSmith, FastAPI, Celery, Redis, GitHub API, Docker</i>	Mar. 2026
- Developed a 6-agent CrewAI code review system with specialized agents for bug detection, security analysis, performance optimization, best practices, automated fixing, and architectural review across code snippets and full GitHub repositories. - Integrated GitHub OAuth and GitHub API workflows to fork repositories, push corrected files, and autonomously create pull requests; used LangSmith for full tracing of prompts, tool calls, latency, and token costs across agent executions. - Built a Dockerized distributed architecture using FastAPI, Celery, Redis, and React/Nginx with asynchronous status polling, enabling snippet reviews in under 60 seconds and full repository reviews within 3–10 minutes.	

TECHNICAL SKILLS

Agentic AI: LangGraph, CrewAI, LangChain, MCP, RAG, Hugging Face Transformers, OpenAI API
LLMOps & Automation: LangSmith, Prompt Engineering, Token Optimization, Cost Tracking, n8n, Make.com
Backend & Infra: FastAPI, Flask, Celery, Redis, Docker, GitHub Actions, AWS (S3, EC2, ECR)
ML & DL: PyTorch, TensorFlow, Scikit-learn, Transfer Learning, Knowledge Distillation, MLflow
Languages & DB & Vector DB: Python, SQL PostgreSQL, MongoDB Weaviate, Pinecone, FAISS

CERTIFICATIONS

Udemy Full Stack Generative and Agentic AI with Python Hitesh Choudhary & Piyush Garg	Apr. 2026
CampusX Crew AI Nitish Singh	Feb. 2026
IBM Machine Learning with Python	Dec. 2025

PUBLICATIONS

SentimentFlow: A Three-Tier Automated Pipeline for Entity-Specific News Sentiment Analysis of Startups Using Zero-Shot Natural Language Inference (Accepted for Publication) Accepted at <i>PTEMS 2026</i>	(Springer Nature)
--	-------------------